

AT-5

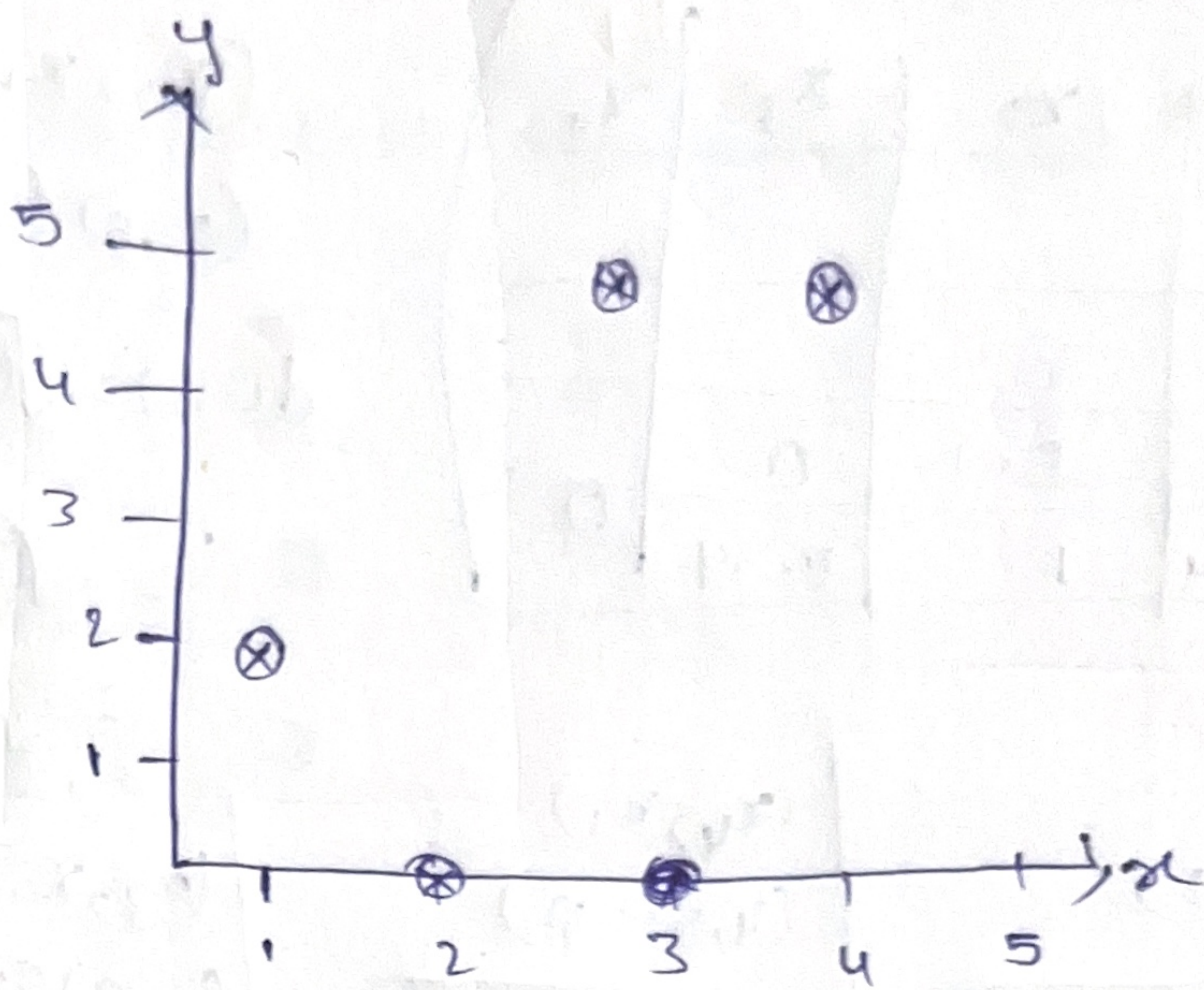
Manhattan distance

$$x_1(1, 2)$$

$$x_2(3, 5)$$

$$x_3(2, 0)$$

$$x_4(4, 5)$$



Dissimilarity Matrices

Manhattan (L_1)

L	x_1	x_2	x_3	x_4
x_1	0			
x_2	5	0		
x_3	3	6	0	
x_4	6	1	7	0

$$(x_2, x_1)$$

$$(3, 5)(1, 2)$$

$$d(i, j) = |x_1 - y_1| + |x_2 - y_2|$$

$$= |3 - 1| + |5 - 2|$$

$$= |2| + |3|$$

$$= 5$$

$$(x_3, x_1)$$

$$(2, 0)(1, 2)$$

$$d(i, j) = |x_1 - y_1| + |x_2 - y_2|$$

$$= |2 - 1| + |0 - 2|$$

$$= 1 + 2$$

$$= 3$$

$$(x_4, x_1)$$

$$(4, 5)(1, 2)$$

$$d(i, j) = |x_1 - y_1| + |x_2 - y_2|$$

$$= |4 - 1| + |5 - 2|$$

$$= 3 + 3$$

$$= 6$$

$$(x_3, x_2)$$

$$(2, 0)(3, 5)$$

$$d = |x_1 - y_1| + |x_2 - y_2|$$

$$= |2 - 3| + |0 - 5|$$

$$= 1 + 5$$

$$(x_4, x_2)$$

$$(4, 5)(3, 5)$$

$$d = |4 - 3| + |5 - 5|$$

$$= 1 + 0$$

$$= 1$$

$$(x_4, x_3)$$

$$(4, 5)(2, 0)$$

$$d = |4 - 2| + |5 - 0|$$

$$= 2 + 5$$

$$= 7$$

Euclidean (L₂)

i	x ₁	x ₂	x ₃	x ₄
x ₁	0			
x ₂	3.61	0		
x ₃	2.24	5.1	0	
x ₄	3.24	1	5.39	0

(x₃, x₁)

(2, 0) (1, 2)

$$d = \sqrt{(2-1)^2 + (0-2)^2}$$

$$= \sqrt{5}$$

(x₄, x₁)

(4, 5) (1, 2)

$$d = \sqrt{(4-1)^2 + (5-2)^2}$$

$$= \sqrt{9+9}$$

$$= \sqrt{18}$$

(x₂, x₁)

(3, 5) (1, 2)

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

$$= \sqrt{(3-1)^2 + (5-2)^2}$$

$$= \sqrt{(2)^2 + (3)^2} = \sqrt{4+9} = \sqrt{13}$$

$$= 3.6$$

(x₃, x₂)

(2, 0) (3, 5)

$$d = \sqrt{(2-3)^2 + (0-5)^2}$$

$$= \sqrt{26}$$

(x₄, x₃)

(4, 5) (2, 0)

$$d = \sqrt{(4-2)^2 + (5-0)^2}$$

$$= \sqrt{29}$$

Supermum

i	x ₁	x ₂	x ₃	x ₄
x ₁	0			
x ₂	3	0		
x ₃	2	5	0	
x ₄	3	1	5	0

(x₂, x₁)

(3, 5) (1, 2)

$$= \max(|x_1 - y_1|, |x_2 - y_2|)$$

$$= \max(2, 3)$$

(x₃, x₁)

(2, 0) (1, 2)

$$= \max(1, 2)$$

(x₄, x₁)

(4, 5) (1, 2)

$$= \max(3, 3)$$

(x₃, x₂)

(2, 0) (3, 5)

$$= \max(1, 5)$$

(x₄, x₂)

(4, 5) (3, 5)

$$= \max(1, 0)$$

(x₄, x₃)

(4, 5) (2, 0)

$$= \max(2, 5)$$

Cosine Similarity

Document	team coach		hockey	baseball	soccer	penalty	score	win	loss	season
1	5	0	3	0	2	0	0	2	0	0
2	3	0	2	0	1	1	0	1	0	1
3	0	7	0	2	1	0	0	3	0	0
4	0	1	0	0	1	2	2	0	3	0

find similarity between Document 1 and 2

$$\cos(d_1, d_2) = (d_1 \cdot d_2) / \|d_1\| \|d_2\|$$

where $\cdot$ indicates vector dot product.

Given

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$\begin{aligned} d_1 \cdot d_2 &= 5 \cdot 3 + 0 \cdot 0 + 3 \cdot 2 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 \\ &= 15 + 6 + 2 + 2 \\ &= 25 \end{aligned}$$

$$\begin{aligned} \|d_1\| &= \sqrt{5 \cdot 5 + 0 \cdot 0 + 3 \cdot 3 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 0 \cdot 0} \\ &= \sqrt{25 + 9 + 4 + 4} \\ &= \sqrt{42} = 6.48 \end{aligned}$$

$$\begin{aligned} \|d_2\| &= \sqrt{3 \cdot 3 + 0 \cdot 0 + 2 \cdot 2 + 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1} \\ &= \sqrt{9 + 4 + 1 + 1 + 1 + 1} \\ &= \sqrt{17} = 4.12 \end{aligned}$$